

**QUESTION ONE (6 Marks)****MULTIPLE CHOICE:** Write the correct alternative on your writing paper.

1. Which of the following gives  $\sqrt{3} - i$  in modulus-argument form 1

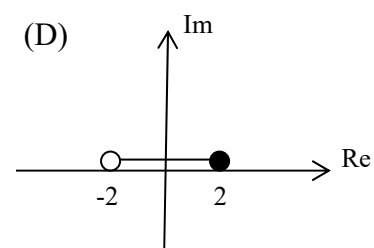
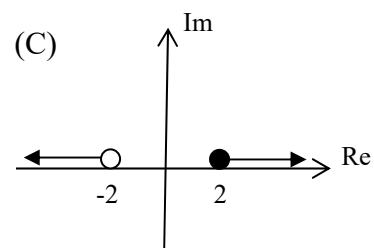
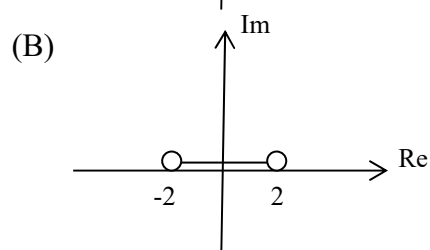
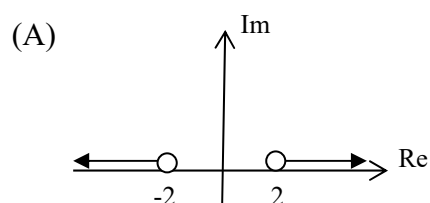
- (A)  $2 \operatorname{cis} \left(-\frac{\pi}{6}\right)$   
(B)  $2 \operatorname{cis} \left(-\frac{2\pi}{3}\right)$   
(C)  $2 \operatorname{cis} \left(\frac{\pi}{6}\right)$   
(D)  $2 \operatorname{cis} \left(\frac{2\pi}{3}\right)$

2. Which of the following represent  $\operatorname{Re}(Z)$  if  $Z = \frac{4+3i}{1-2i}$  1

- (A)  $-\frac{10}{3}$   
(B)  $-\frac{2}{5}$   
(C) 4  
(D) -2

3. Which of the following represent the locus of  $Z$  satisfying the condition 1

$$\arg\left(\frac{Z-2}{Z+2}\right) = 0$$



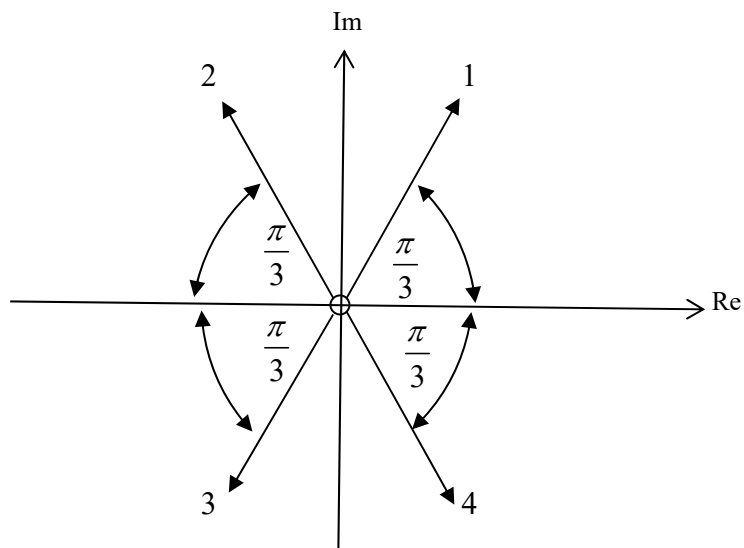
4. Which of the following are the square roots of  $-16 + 30i$ ? 1

- (A)  $\pm(3 - 5i)$
- (B)  $\pm(5 - 3i)$
- (C)  $\pm(5 + 3i)$
- (D)  $\pm(3 + 5i)$

5. Which is the simplified form of  $i^{2011}$ ? 1

- (A)  $i$
- (B)  $-1$
- (C)  $-i$
- (D)  $1$

6. Which ray represents the locus  $\arg(-Z) = \frac{\pi}{3}$ ? 1



- (A) 1
- (B) 2
- (C) 3
- (D) 4

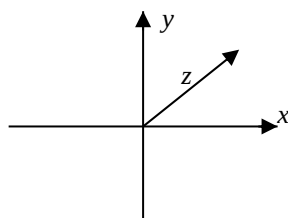
**QUESTION TWO (6Marks) Start a new page**

- (a) Let  $z = 4 - i$ . 3
- (i) Find  $z^2$  in the form  $x + iy$ .
- (ii) Find  $z + 2\bar{z}$  in the form  $x + iy$ .
- (iii) Find  $\frac{i}{z}$  in the form  $x + iy$ .
- (b) Show that  $(-\sqrt{3} - i)^6$  is a real number. 3

**QUESTION THREE (12 Marks) Start a new page**

- (a) Let  $A$  be the point representing  $-1 - i$  and  $B$  be the point representing  $3 + i$  in the Argand plane. Draw a sketch of the locus of the points  $z$ , so that  $|z - A| \leq |z - B|$ . 2

(b)



The diagram above shows a vector representing the complex number  $z$ . Copy the diagram and on it show vectors representing the complex numbers  $iz$  and  $z - iz$ .

3

- (c) The points  $O, I, Z$  and  $P$  on the Argand diagram represent the complex numbers  $0, 1, z$  and  $z + 1$  respectively, where  $z = \cos \theta + i \sin \theta$  is any complex number of modulus 1 and  $0 < \theta < \pi$ . 3
- (i) Explain why  $OIZP$  is a rhombus.
- (ii) Show that  $\frac{z-1}{z+1}$  is purely imaginary.
- (c) (i) Express  $(-1 + \sqrt{3}i)(1 + i)$  in the form  $a + ib$  4
- (ii) Hence, or otherwise, find the exact value of  $\cos \frac{11\pi}{12}$

**QUESTION FOUR (16Marks) Start a new page**

- (a) (i) Write down the general solution of  $\tan 4\theta = 1$ . **9**
- (ii) Use De Moivre's Theorem to express  $\cos 4\theta$  and  $\sin 4\theta$  in terms of  $\cos \theta$  and  $\sin \theta$ .
- (iii) Hence show that  $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$ .
- (iv) Hence or otherwise, find the roots of the equation  $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ .
- (v) Hence show that  $\tan \frac{\pi}{16} + \tan \frac{5\pi}{16} = \tan \frac{3\pi}{16} + \tan \frac{7\pi}{16} - 4$ .
- (b) (i) Find all solutions to the equation  $z^6 = 1$  in the form  $x + iy$ . **7**
- (ii) If  $\omega$  is the non real solution to the equation  $z^6 = 1$ , show that  $\omega^4 + \omega^2 = -1$ .
- (iii) By choosing on particular value of  $\omega$ , explain with the aid of a diagram why  $\omega^4 + \omega^2 = -1$ .

**END OF PAPER**

Question One

1.  $z = \sqrt{3} - i$

$$|z| = \sqrt{(\sqrt{3})^2 + 1^2}$$

$$= 2$$

$$\arg(z) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= -\frac{\pi}{6}$$

$$\therefore \sqrt{3} - i = 2 \operatorname{cis}\left(-\frac{\pi}{6}\right) \quad (\text{A})$$

2.  $z = \frac{4+3i}{1-2i}$

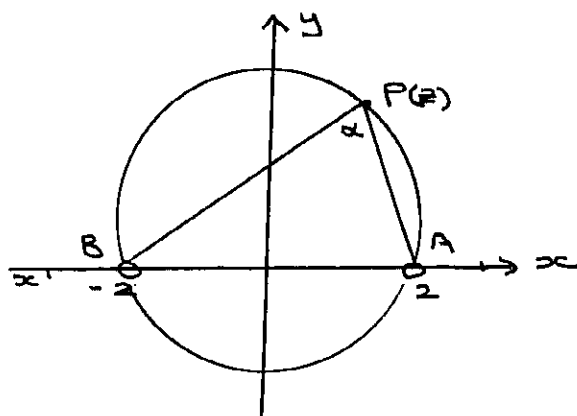
$$= \frac{(4+3i)}{(1-2i)} \times \frac{(1+2i)}{(1+2i)}$$

$$= \frac{4+11i-6}{1+4}$$

$$= -\frac{2}{5} + \frac{11}{5}i$$

$$\therefore \operatorname{Re}(z) = -\frac{2}{5} \quad (\text{B})$$

3.

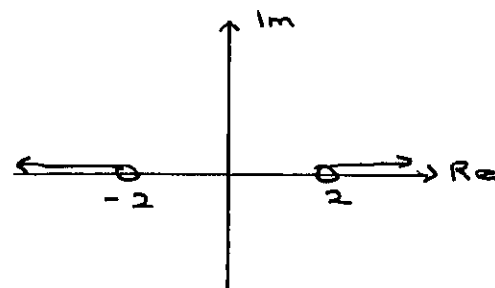


$$\arg\left(\frac{z-2}{z+2}\right) = \arg(z-2) - \arg(z+2)$$

$$= 0$$

$$\text{If } \alpha = 0, \hat{PAx} = \hat{PBx}$$

$\therefore$  locus of P is the rays Ax, and Bx'



(A)

4. Let  $x+iy = \sqrt{-16+30i}$

squaring both sides

$$(x+iy)^2 = -16+30i$$

$$x^2 - y^2 + 2xyi = -16 + 30i$$

equating real and imaginary parts

$$x^2 - y^2 = -16$$

$$2xy = 30$$

$$x = \pm 3$$

$$y = \pm 5$$

 $\therefore$  The 2 square roots are

$$3+5i \text{ and } -3-5i$$

$$\text{that is } \pm(3+5i) \quad (\text{D})$$

5.  $\frac{2011}{4} = 502 \frac{3}{4}$

$$\therefore L^{2011} = L^3$$

$$= -i \quad (\text{C})$$

b. (C)

## Question Two

a) i) Let  $z = 4 - i$

$$\begin{aligned} z^2 &= (4 - i)^2 \\ &= 16 - 8i + i^2 \\ &= 15 - 8i \end{aligned}$$

$$\begin{aligned} \text{ii) } z + 2\bar{z} &= 4 - i + 2(4 + i) \\ &= 4 - i + 8 + 2i \\ &= 12 + i \end{aligned}$$

$$\begin{aligned} \text{iii) } \frac{1}{z} &= \frac{1}{4 - i} \\ &= \frac{i}{(4 - i) \times (4 + i)} \\ &= \frac{-1 + 4i}{17} \\ &= -\frac{1}{17} + \frac{4}{17}i \end{aligned}$$

b) Let  $z = -\sqrt{3} - i$

$$\begin{aligned} |z| &= \sqrt{(\sqrt{3})^2 + 1^2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \arg(z) &= \tan^{-1} \frac{1}{\sqrt{3}} \\ &= -\frac{5\pi}{6} \end{aligned}$$

$$\therefore -\sqrt{3} - i = 2 \operatorname{cis}\left(-\frac{5\pi}{6}\right)$$

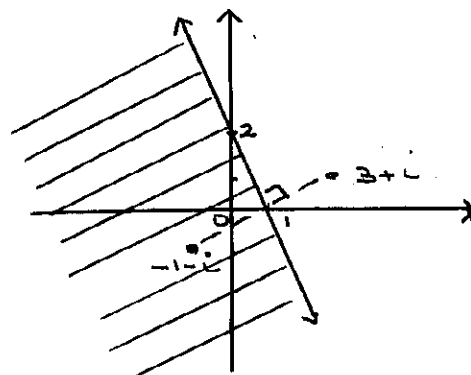
Using De Moivre's Theorem

$$\begin{aligned} (-\sqrt{3} - i)^6 &= \left[2 \operatorname{cis}\left(-\frac{5\pi}{6}\right)\right]^6 \\ &= 2^6 \operatorname{cis}(-\pi) \\ &= -64 \end{aligned}$$

$\therefore (-\sqrt{3} - i)^6$  is a real number

## Question Three

a)



$$(-1, -1) \quad (3, 1)$$

$$\begin{aligned} \text{midpoint} &= \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left( \frac{-1 + 3}{2}, \frac{-1 + 1}{2} \right) \\ &= (1, 0) \end{aligned}$$

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{1 + 1}{3 + 1} \\ &= \frac{1}{2} \end{aligned}$$

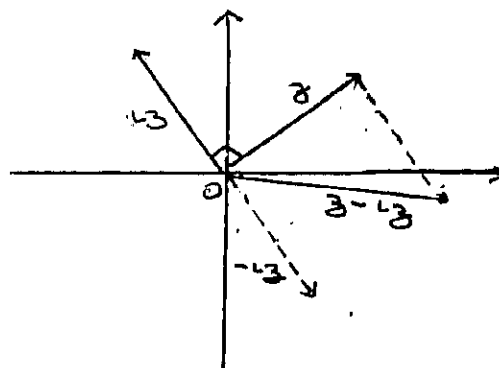
$\therefore$  Required gradient = -2  
as  $m_1 m_2 = -1$  for perpendicular lines

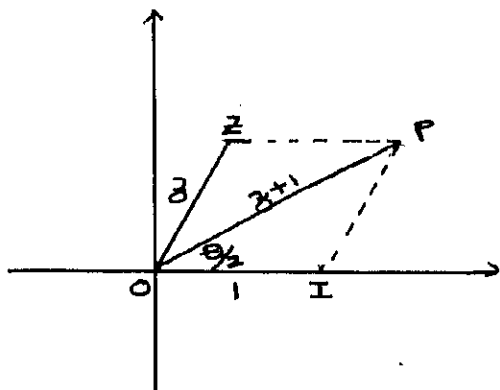
$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 0 &= -2(x - 1) \\ y &= -2x + 2 \end{aligned}$$

b) Let  $z = x + iy$

$$iz = -y + ix$$

$\therefore z$  is rotated  $90^\circ$  in an anticlockwise direction





$$i) \quad z = \cos \theta + i \sin \theta$$

$$|z| = 1$$

OIPZ is a parallelogram

$$OI = OZ \\ = 1$$

$\therefore$  as the adjacent sides are equal OIPZ is a rhombus

ii)  $OP \perp ZI$  as the diagonals bisect at right angles

OP represents  $z+1$

OI represents  $z-1$

$$\therefore z-1 = k i (z+1)$$

$\therefore \frac{z-1}{z+1}$  is purely

imaginary

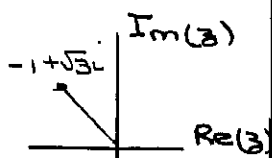
$$c) \quad (-1 + \sqrt{3}i)(1+i) = -1 - i + \sqrt{3}i - \sqrt{3} \\ = -(1+\sqrt{3}) + (\sqrt{3}-1)i$$

$$\text{Let } z = -1 + \sqrt{3}i$$

$$|z| = \sqrt{1^2 + (\sqrt{3})^2} \\ = 2$$

$$\arg z = \pi - \tan^{-1}(\sqrt{3}) \\ = \frac{2\pi}{3}$$

$$\therefore -1 + \sqrt{3}i = 2 \cos\left(\frac{2\pi}{3}\right)$$



$$w = (1+i)$$

$$|w| = \sqrt{1^2 + 1^2} \\ = \sqrt{2}$$

$$\arg(w) = \tan^{-1} 1 \\ = \frac{\pi}{4}$$

$$\therefore 1+i = \sqrt{2} \cos\left(\frac{\pi}{4}\right)$$

$$\therefore (-1 + \sqrt{3}i)(1+i) = 2 \cos\left(\frac{2\pi}{3}\right) \sqrt{2} \cos\left(\frac{\pi}{4}\right) \\ = 2\sqrt{2} \cos\left(\frac{2\pi}{3} + \frac{\pi}{4}\right) \\ = 2\sqrt{2} \cos\left(\frac{11\pi}{12}\right)$$

Equating real parts

$$2\sqrt{2} \cos \frac{11\pi}{12} = -(1+\sqrt{3})$$

$$\cos \frac{11\pi}{12} = \frac{-(1+\sqrt{3})}{2\sqrt{2}}$$

#### Question Four

$$a) \quad i) \quad \tan 4\theta = 1$$

$$\tan \frac{\pi}{4} = 1$$

$$\therefore 4\theta = n\pi + \frac{\pi}{4}$$

$$\theta = \frac{n\pi}{4} + \frac{\pi}{16} \quad n \in \mathbb{Z}$$

ii)  $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$  by De Moivre's Theorem

$$\begin{aligned} \text{L.H.S} &= (\cos \theta + i \sin \theta)^4 \\ &= \cos^4 \theta + 4\cos^3 \theta (i \sin \theta) + 6\cos^2 \theta (i \sin \theta)^2 + 4\cos \theta (i \sin \theta)^3 + (i \sin \theta)^4 \\ &= \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta + i(4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta) \end{aligned}$$

Equating real parts

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta \quad \text{--- ①}$$

Equating imaginary parts

$$\sin 4\theta = 4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta \quad \text{--- ②}$$

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

$$= \frac{4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta}{\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

$$= \frac{4\tan \theta - 4\tan^3 \theta}{1 - 6\tan^2 \theta + \tan^4 \theta}$$

(iii) Now L.H.S = 1 when  $\tan 4\theta = 1$

$$\text{Let } x = \tan \theta$$

$$\therefore 1 = \frac{4x - 4x^3}{1 - x^2 + x^4}$$

$$x^4 - x^2 + 1 = 4x - 4x^3$$

$$x^4 + 4x^3 - x^2 - 4x + 1 = 0 \quad \text{as required}$$

$$\therefore \tan 4\theta = 1$$

$$4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4} \quad \text{NB from (a)}$$

$$\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16} \quad \theta = \frac{n\pi}{4} + \frac{\pi}{16} \text{ for } n=0,1,2,3$$

$$\therefore x = \tan \theta$$

$$x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$$

$$\therefore x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{7\pi}{16}, -\tan \frac{3\pi}{16}$$

as  $\tan \theta$  is  
negative in the  
second quadrant

$$(iv) \text{ the sum of the roots} = \frac{-b}{a} \\ = -4$$

$$\therefore \tan \frac{\pi}{16} + \tan \frac{5\pi}{16} - \tan \frac{7\pi}{16} - \tan \frac{3\pi}{16} = -4$$

$$\therefore \tan \frac{\pi}{16} + \tan \frac{5\pi}{16} = \tan \frac{3\pi}{16} + \tan \frac{7\pi}{16} - 4 \\ \text{as required.}$$

b) u)

$$z^6 = 1$$

$$z^6 = \cos(2k\pi) \quad k \in \mathbb{Z}$$

$$z = \cos\left(\frac{2k\pi}{6}\right) \text{ by De Moivre's Theorem} \\ = \cos\left(\frac{k\pi}{3}\right)$$

For  $k = 0$

$$z = \cos 0$$

$$z = 1$$

$k = 1$

$$z = \cos\left(\frac{\pi}{3}\right)$$

$$= \cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right)$$

$$\therefore z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$k = 2$

$$z = \cos\left(\frac{2\pi}{3}\right)$$

$$= \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)$$

$$z = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$k = 3$

$$z = \cos \pi$$

$$= \cos \pi + i \sin \pi$$

$$= -1$$

$k = -1$

$$z = \cos\left(-\frac{\pi}{3}\right)$$

$$= \cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right)$$

$$= \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$k = -2$

$$z = \cos\left(-\frac{2\pi}{3}\right)$$

$$z = \cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right)$$

$$= -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$\therefore$  The solutions are  $\pm 1, \frac{1}{2} \pm \frac{\sqrt{3}}{2}i, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$

$$(ii) \quad z^6 - 1 = 0$$

$$(z^3 - 1)(z^3 + 1) = 0$$

$$(z - 1)(z^2 + z + 1)(z + 1)(z^2 - z + 1) = 0$$

The 2 real roots are given by  $(z - 1)(z + 1)$

The 4 non real roots are given by  $(z^2 + z + 1)(z^2 - z + 1)$

Let  $\omega$  be the complex roots

$$(\omega^2 + \omega + 1)(\omega^2 - \omega + 1) = 0$$

$$\omega^4 - \omega^3 + \omega^2 + \omega^3 - \omega^2 + \omega + \omega^2 - \omega + 1 = 0$$

$$\omega^4 + \omega^2 + 1 = 0$$

$\therefore \omega^4 + \omega^2 = -1$  as required

(iii) From (i) let  $\omega = \cos\left(\frac{\pi}{3}\right)$

$\omega^4 = \cos\left(\frac{4\pi}{3}\right)$  by De Moivre's Theorem

$$\omega^2 = \cos\left(\frac{2\pi}{3}\right)$$

$$\begin{aligned}\cos\frac{4\pi}{3} + \cos\frac{2\pi}{3} &= \cos\left(\frac{4\pi}{3}\right) + i\sin\left(\frac{4\pi}{3}\right) + \cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) \\ &= -\frac{1}{2} + \frac{\sqrt{3}}{2}i + -\frac{1}{2} - \frac{\sqrt{3}}{2}i \\ &= -1\end{aligned}$$

